

Problem 4

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Problem 1 Determine the following limits. Use L'Hôpital's Rule if applicable.

(a) $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}}$

(b) $\lim_{x \rightarrow -\infty} x^2 e^x$

(c) $\lim_{x \rightarrow \infty} \frac{1}{x^x}$

(d) $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x$

(e) $\lim_{\theta \rightarrow 0^+} (\sin(\theta))^{\tan(\theta)}$